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The Oort Cloud as an External-Field-Effect Instrument: a Conditional Joint Signature with Gaia DR4

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Abstract

Of everything a star owns, only its comet cloud extends into the regime where the acceleration scale $a_0 = \kappa c \sqrt{G\rho\Lambda} = 9.3619 \times 10^{-11} \text{ m s}^{-2}$ becomes dynamically relevant: the MOND radius of a solar-mass star is $r_M \approx 7,960 \text{ AU}$ (canonical footing; 7,250 alt), and Oort-cloud comets are stored at $10^3\text{--}10^5 \text{ AU}$. We price this channel honestly, in both directions. **Three results.** (1) At solar-neighbourhood external fields the external-field effect (EFE) removes almost the entire deep-MOND zone: the cloud is Newtonian inside $\sim r_M$ and anisotropic quasi-Newtonian beyond $\sim 0.88 r_M$, governed by the same committed response tensor ($B = 1.4732$, $B_- = 1.2598$) as the hash-frozen Gaia DR4 wide-binary registration. A uniform ν -boost of gravity nearly self-cancels in comet-injection physics (the energy spike shifts only as $G_{\text{eff}}^{(1/7)}$, $\approx 6\%$); **the surviving signal is directional** — a $\sim 7\%$ aphelion-direction modulation of the spike position and a $\sim 17\%$ modulation of the injection rate, keyed to the galactic external-field direction. (2) The *exo*-Oort version of this test — KIC 8462852 (“Tabby’s Star”) read through its dips — is **dead by 6.7 orders of magnitude**: the inference chain from dip statistics to gravity leaks a factor $\sim 9 \times 10^5$ of model slack against the ~ 0.17 signal, priced arrow by arrow. (3) In the full relativistic completion the extra structure is *silent* at cloud scales (the AeST mass term’s oscillation radius is $\geq 23 \text{ kpc}$ at the pinned Q_0 band; the a_0 -bump term bounds out at $\geq 280 \text{ kpc}$; the derived $a_0(z)$ is flat to 5×10^{-7} over the injection history) — but the pressure promotion $\mathcal{A}(Q) = a_0^2(Q)$ makes a_0 local, so the comet anisotropy becomes ν_0 -dependent: **16.7% at the charge floor, 12.1% at the ceiling** (steady-state-continuity reading), co-varying with the DR4 wide-binary γ_v . Constant- a_0 MOND predicts a fixed 17%; this framework predicts a **correlated pair** — one dimensionless charge moves the comet statistics and the wide-binary boost together. This note prices a door; it does not claim a detection, and it freezes nothing.

1. The gravitational geography of a comet cloud

For a star of mass M the framework’s transition radius is $r_M = \sqrt{GM/a_0}$:

object	footing	r_M [AU]	EFE-dominated beyond [AU]
Sun	canonical	7,960	~7,000
Sun	alt	7,250	~6,400
KIC 8462852 (1.43 M)	canonical	9,520	~8,400

Planets, asteroid belts, and Kuiper belts all sit deep inside the Newtonian zone (the framework’s Newtonian residual is $e^{\sqrt{y}}$: $\sim 10^{-3457}$ at 1 AU). The Oort reservoir straddles r_M — but the observed solar-neighbourhood external field, $x_{\text{ext}} = g_{\text{ext}}/a_0 = 1.9$, dominates the internal field once $y_{\text{int}} < y_{\text{ext}N} = 1.29$, i.e. beyond $0.88 r_M$. The isolated deep-MOND zone is a sliver; what a comet cloud actually experiences outside $\sim r_M$ is quasi-Newtonian gravity with the anisotropic linear-response tensor of the committed AQUAL-EFE solve: an effective coupling 1.4732 G along the external-field direction and 1.2598 G across it.

2. The self-cancellation, and the signal that survives

Comet injection is a competition between the star’s binding and the galactic tide: the tide walks perihelia into the planetary loss cone, and the balance sets the observed energy spike of long-period comets. Both sides of that balance carry the ν -boost, so a *uniform* enhancement largely cancels: at leading tide-scaling order the spike position shifts only as $G_{\text{eff}}^{1/7}$ ($\approx +6\%$ at $\nu = 1.4732$). What does not cancel is the *anisotropy*: with injection efficiency scaling as the directional G_{eff} , the tensor imprints

- a **~7%** modulation of the spike position with aphelion direction ($\propto (B_-/B_+)^{3/7}$), and
- a **~17%** modulation of the injection rate ($\propto B_-/B_+ = 1.169$),

organized relative to the galactic external-field direction. Newtonian tide theory already predicts (and the long-period-comet catalogs show) galactic-latitude structure in aphelia directions; the framework modifies its *amplitude* by these factors. The $2/7$ and $3/7$ exponents are leading-order tide scalings, stated not derived — a named owed item.

3. Why the exo version is dead: KIC 8462852, priced arrow by arrow

Tabby’s Star’s dips, if cometary, sample an exo-Oort cloud — the only way “shadows” ever touch a_0 physics. The chain, priced:

inference arrow	model slack
dips $\rightarrow$ comet interpretation (dust family competes; not established)	$\times 10$
dip depth $\rightarrow$ swarm mass $\rightarrow$ comet number	$\times 30$
comet rate $\rightarrow$ injection rate (unknown cloud population $N(a)$)	$\times 300$
injection rate $\rightarrow$ cloud dynamics (unknown stellar history/flybys)	$\times 10$

inference arrow	model slack
total slack vs the ~0.17 anisotropy signal	$\sim 9 \times 10^5$ — 6.7 orders

The star’s 880-AU companion is itself gravity-boring ($y = 117$; Newtonian residual $\sim 2 \times 10^{-5}$). Verdict: dead as a gravity instrument — a dust mystery pointed away from gravity. Any exo-Oort gravity test inherits this chain; the solar cloud does not.

4. The honest cousin: our own Oort spike

The observed long-period-comet energy spike sits at $1/a \approx (3-5) \times 10^{-5} \text{ AU}^{-1}$ ($a \approx 20,000-33,000 \text{ AU}$), with hundreds of quality orbits in existing catalogs. At the spike, $y_{\text{int}} \approx 0.16 \ll y_{\text{extN}} = 1.29$: the physics is EFE-dominated *exactly as in the wide-binary problem*, at ten times the separation. A comet-aphelia anisotropy analysis is therefore a **free cross-check of the same tensor the DR4 registration tests**, on data that already exist.

5. The full-completion layer: what AeST adds and what it silences

Run through the complete action (AeST scaffold with the pressure promotion), every additional structure is bounded silent at cloud scales: the mass term $\mu = \sqrt{(2_2/(2-K_B))} \cdot \mathcal{Q}_0$ has its oscillation radius $r_C = (r_M \mu^{-2})^{1/3} \geq 23 \text{ kpc}$ at the pinned $\mathcal{Q}_0$ band; the a_0 -bump cluster response bounds out at $\geq 280 \text{ kpc}$ even at gate maximum; the derived $a_0(z)$ is flat to 5×10^{-7} over the Gyr injection history. The kernel-level prediction of §§2 therefore *is* the framework’s prediction. The completion adds exactly one thing: because $\mathcal{A}(\mathcal{Q}) = a_0^2(\mathcal{Q})$ is local, the solar-circle charge flow suppresses the local a_0 (steady-state-continuity reading), and the anisotropy becomes charge-dependent:

ν_0 (the framework’s one free dimensionless charge)	local a_0 ratio S	rate anisotropy
floor (2.14×10^{-5})	0.977	16.7%
ceiling (1.77×10^{-4})	0.599	12.1%

The same suppression moves the DR4 wide-binary γ_v (the registered band’s conditional ν_0 -meter). **The framework therefore predicts a correlated pair**: the comet anisotropy and the wide-binary boost move together under one parameter, in a computable ratio. Constant- a_0 MOND predicts a fixed 17% with no partner observable. That correlation — not either number alone — is the signature unique to this framework.

6. What this does not establish

- **No detection is claimed and nothing is frozen.** This note prices a channel; the DR4 registration is untouched and this analysis is not part of it.
- The tide-scaling exponents ($2/7$, $3/7$) are leading-order and stated, not derived; the honest next step derives them and the loss-cone geometry properly.
- The known Newtonian-tide anisotropy must be modeled and subtracted before a 7–17% modification of its amplitude is testable; whether $\sim 10^2-10^3$ catalog orbits give the statistical power is an owed analysis, not assumed.

- Prior art exists on Milgromian Oort-cloud dynamics (Paučo & Klačka 2016, Sedna and the comet cloud in Milgromian dynamics) and is flagged LITERATURE-UNVERIFIED here: it must be read and credited in detail before any continuation.
- $\kappa = \frac{1}{2}$ remains fitted, not derived (measured 0.551 ± 0.043); both footings are carried in §§\$1.

Reproducibility

All numbers from one committed script at <https://github.com/carlzimmerman/zimmerman-formula>:

- `real_research/tabby_exo_oort_a0_2026.py` — 13 checks: the geography (both footings), the self-cancellation and anisotropy scalings, the Tabby chain pricing, the solar-spike regime, and the full-AeST layer (mass-term, bump, $a_0(z)$ bounds; the ν_0 co-variance table).
- `prep_2026/gaia_dr4_prep/aqual_efe_full_solve_2026.py` — the committed response tensor.
- `nbody_2026/stage61_drain_reading_fork_2026.py` — the continuity reading and the local- a_0 suppression behind §§\$5.

Prior art and attribution. AeST is Skordis & Zlošnik, PRL **127** 161302; the kernel is Milgrom & Sanders 2008, ApJ **678** 131, Eq. (13) at $\alpha = \frac{1}{2}$; Oort-cloud injection theory is classical (Oort 1950; Heisler & Tremaine 1986); Milgromian Oort-cloud dynamics has prior art in Paučo & Klačka 2016 (to be verified and credited in detail). The framework’s contribution is the a_0 normalisation, the local- a_0 structure, the committed EFE tensor, and the ν_0 -correlated-pair prediction of §§\$5.